

Dynamic Stochastic General Equilibrium and Business Cycles

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CHAPTER 5

Optimal Monetary Policy

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HANDOUT OBJECTIVES

- Incorporating monetary and fiscal policy in a business cycle framework;
- Evaluating the effectiveness of policy using a loss function;
- Performing a welfare ranking of alternative macroeconomic policies;

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1. A Model for Policy Analysis

We develop here a very standard DSGE model with both real and nominal frictions. This is a New Keynesian model with capital services subject to adjustment costs and is very close to the framework of [Smets and Wouters \(2007\)](#) which has become the workhouse for policy experiments. The introduction of new keynesian features allows to have macroeconomic policies effective in stabilising the economy. In this lecture, we more specifically focus on fiscal and monetary policies.

1.1 Households

Our economy is populated by identical and infinite living households distributed over a continuum $j \in [0, 1]$. The representative household utility function reads as follows:

$$\mathcal{U}(C_{jt}, H_{jt}) = \frac{C_{jt}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt}^{1+\sigma_H}}{1+\sigma_H} \quad (1)$$

Utility is increasing in consumption C_{jt} and decreasing in labour H_{jt} where σ_C and σ_L are curvature parameters. Households in this economy maximise the expected value of their utility:

$$\max_{\{C_{jt}, H_{jt}, B_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \mathcal{U}(C_{jt+\tau}, H_{jt+\tau}), \quad (2)$$

where β is the discount factor and ε_t^C is an AR(1) time-preference shock (i.e. a demand shock). Households face an intertemporal problem, they find the value of consumption C_{jt} , hours worked H_{jt} and real bonds B_{jt} which delivers the highest value of utility. They face the following budget constraint which binds every period:

$$C_t + B_t = \frac{r_{t-1}}{\pi_t} B_{t-1} + \Pi_t + w_t H_t \quad (3)$$

where w_t is the real wage and Π_t are real profits from monopolistic competition.

The household must solve the following problem:

$$\begin{aligned} \max_{\{C_t, H_t, B_t\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \varepsilon_{t+\tau}^C \left\{ \frac{C_{jt+\tau}^{1-\sigma_C}}{1-\sigma_C} - \chi \frac{H_{jt+\tau}^{1+\sigma_H}}{1+\sigma_H} \right. \\ \left. + \lambda_{t+\tau}^c \left[\frac{r_{t-1+\tau}}{\pi_{t+\tau}} B_{t-1+\tau} + w_{t+\tau} H_{t+\tau} + \Pi_{t+\tau} - C_{t+\tau} - B_{t+\tau} \right] \right\} \end{aligned}$$

First order conditions are given by:

$$\varepsilon_t^C \chi H_t^\varphi C_t = w_t, \quad (4)$$

$$\mathbb{E}_t \left\{ \beta \left(\frac{C_t}{C_{t+1}} \right)^{\sigma_C} \frac{1}{\pi_{t+1}} \frac{\varepsilon_{t+1}^C}{\varepsilon_t^C} \right\} = \frac{1}{r_t}. \quad (5)$$

1.2 Firms

To introduce a monopolistic competition, the production process of goods is divided between two types of firms: intermediate and final firms. Intermediate firms (or wholesale goods producers) produce different types of goods which are imperfect substitutes (at the origin of the monopolistic competition). Final firms (or retailers) produce an homogenous good by combining intermediate goods.

1.2.1 Final sector

The final good producers are retailers, they buy intermediate goods Y_{it} , package them into Y_t , and sell the final good to households. In equilibrium on the good market, Y_t also denotes the aggregate demand for goods from households. On a perfectly competitive market, final producers maximize profits:

$$P_t Y_t - \int_0^1 p_{it} y_{it} di,$$

subject to a supply constraint:

$$Y_t = \left(\int_0^1 y_{it}^{(\epsilon-1)/\epsilon} di \right)^{\epsilon/(\epsilon-1)}. \quad (6)$$

This supply constraint implies that final producers have a technology which aggregate non-perfectly substitutable goods. This imperfect substitutability between all types of varieties i is driven by the monopolistic competition on the intermediate good market. Each good i is an imperfect substitute of degree $\epsilon > 1$, allowing intermediate firms to gain positive profits through a gap between their selling and producing price. The intensity of the monopolistic competition is driven by $\epsilon/(\epsilon-1)$ the mark-up over the producing price of intermediate firms.

Finally, we find the intermediate demand function for each variety i is:

$$Y_{it} = \left(\frac{P_{it}}{P_t} \right)^{-\epsilon} Y_t^d, \quad \forall i \quad (7)$$

1.2.2 Intermediate sector

Firms are homogeneous and distributed on a unitary interval $i \in [0, 1]$ and have the following production technology:

$$y_{it} = \varepsilon_t^A h_{it} \quad (8)$$

where y_{it} is the production, h_{it} is an input and ε_t^A denotes the productivity shock process.

Intermediate goods producers solve a two-stage problem. In the first stage, taken the input price w_t as given, firms rent input h_{it} in a perfectly competitive factor market in order to minimize costs subject to the production constraint 8.

The first one can be expressed as a profit maximisation problem:

$$\max_{Y_{it}, K_{it-1}, H_{it}} mc_{it} y_{it} - w_t H_{it} + \lambda_t [\varepsilon_t^A h_{it} - y_{it}]$$

where mc_{it} denotes the real marginal cost of producing one additional good. First order conditions leads to the expression of the real marginal cost:

$$mc_t = \frac{w_t}{\varepsilon_t^A} \quad (9)$$

In the second stage problem, the firms decides its selling price on a Rotemberg price setting. We define the rotemberg price adjustment cost:

$$ac_{it} = \frac{\kappa}{2} \left(\frac{p_{it}}{p_{it-1}} - 1 \right)^2 Y_t$$

where $\chi > 0$ is the price stickiness parameter and Y_t is aggregate output. Firms must pay a cost when adjusting the price that is proportional to the total sold quantity Y_t .

The profit maximisation becomes dynamics because of the ajustment costs over prices. In a monopolistic competition setting, firms face the following individual demand for goods: $y_{it} = (p_{it}/P_t)^{-\epsilon} Y_t$. The problem faced by firms is given by:

$$\max_{\{p_{it}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \Lambda_{t,t+\tau} \left(y_{it+\tau} \frac{p_{it+\tau}}{P_{t+\tau}} - \varepsilon_{t+\tau}^P mc_{t+\tau} - ac_{it+\tau} Y_{t+\tau} \right)$$

where $\Lambda_{t,t+\tau} = \beta^\tau (C_t/C_{t+\tau})^{\sigma_C}$ is the household discount factor, p_{it} is the individual price set by firm i , P_t is the aggregate price which sets in real terms the problem of firms, ε_t^P is a cost push shock.

Replacing the demand function and the technology:

$$\max_{\{p_{it}\}} \mathbb{E}_t \sum_{\tau=0}^{\infty} \Lambda_{t,t+\tau} \left(\left(\frac{p_{it+\tau}}{P_{t+\tau}} \right)^{1-\epsilon} Y_{t+\tau} - mc_{t+\tau} \left(\frac{p_{it+\tau}}{P_{t+\tau}} \right)^{-\epsilon} Y_{t+\tau} - ac_{it+\tau} Y_{t+\tau} \right)$$

Once the optimization in p_{it} is done, we can aggregate firms to simplify the expression $p_{it} = P_t$. The optimal price reads as follows:

$$(\epsilon - 1) - \epsilon \varepsilon_t^P mc_t + \kappa (\pi_t - 1) \pi_t - \beta \mathbb{E}_t \left(\frac{C_t}{C_{t+1}} \right)^{\sigma_C} \kappa (\pi_{t+1} - 1) \pi_{t+1} \frac{Y_{t+1}}{Y_t} = 0$$

1.2.3 Monetary policy and aggregation

To close the model, the monetary policy authority sets its interest rate according to a standard Taylor rule:

$$r_t = r_{t-1}^\rho \left[\bar{r} \left(\frac{\pi_t}{\bar{\pi}} \right)^{\phi_\pi} \left(\frac{y_t}{\bar{y}} \right)^{\phi_y} \right]^{1-\rho} \quad (10)$$

here R_t is the nominal interest rate, π_t is the inflation rate and ε_t^R is an AR(1) monetary policy shock. Finally, parameters $\bar{R}$ and $\bar{\pi}$ are long term values for the interest rate and the inflation rate. The central bank reacts to the deviation of the inflation rate from its steady state value in a proportion ϕ_π , the central bank also smooths her rate at a degree ρ .

The resource constraint is given by:

$$Y_t = C_t + \frac{\kappa}{2} (\pi_t - 1)^2 Y_t \quad (11)$$

2. Summary of the model

The set of equations of the model reads as follows:

$$\varepsilon_t^C \chi H_t^\varphi C_t = w_t, \quad (12)$$

$$\mathbb{E}_t \left\{ \beta \left(\frac{C_t}{C_{t+1}} \right)^{\sigma_C} \frac{1}{\pi_{t+1}} \frac{\varepsilon_{t+1}^C}{\varepsilon_t^C} \right\} = \frac{1}{r_t}, \quad (13)$$

$$Y_t = \varepsilon_t^A H_t, \quad (14)$$

$$mc_t = \frac{w_t}{\varepsilon_t^A}, \quad (15)$$

$$(\epsilon - 1) - \epsilon \varepsilon_t^P mc_t + \kappa (\pi_t - 1) \pi_t - \beta \mathbb{E}_t \left(\frac{C_t}{C_{t+1}} \right)^{\sigma_C} \kappa (\pi_{t+1} - 1) \pi_{t+1} \frac{Y_{t+1}}{Y_t} = 0, \quad (16)$$

$$r_t = r_{t-1}^\rho \left[\bar{r} \left(\frac{\pi_t}{\bar{\pi}} \right)^{\phi_\pi} \left(\frac{y_t}{\bar{y}} \right)^{\phi_y} \right]^{1-\rho}, \quad (17)$$

$$Y_t = C_t + \frac{\kappa}{2} (\pi_t - 1)^2 Y_t, \quad (18)$$

$$\log(\varepsilon_t^A) = \rho_A \log(\varepsilon_{t-1}^A) + \eta_t^A, \quad (19)$$

$$\log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C, \quad (20)$$

$$\log(\varepsilon_t^P) = \rho_P \log(\varepsilon_{t-1}^P) + \eta_t^P \kappa / (\epsilon - 1). \quad (21)$$

Finally, the steady state is straightforward to obtain as long as we normalize to one the hours supply:

$$\bar{Y} = \bar{C} = \bar{H} = \bar{\pi} = 1, \quad (22)$$

$$\chi = \frac{(\epsilon - 1)}{\bar{H}^{1+\varphi_\epsilon}} = \frac{(\epsilon - 1)}{\epsilon}. \quad (23)$$

and we set the rest of parameters as follows: $\beta = 0.994$, $\sigma_C = 1.5$, $\sigma_H = 2$, $\kappa = 150$, $\epsilon = 6$, $\rho = 0.8$, $\phi_\pi = 1.5$ and $\phi_y = 0.125$. The persistence of shocks ρ_A , ρ_C and ρ_P and their standard deviations of shocks σ_A , σ_C and σ_P are taken from [Smets and Wouters \(2007\)](#).

3. Optimal policy

Up to now, we have discussed the implementation of monetary policy in terms of a simple rule for the interest rate calibrated without any concern for macroeconomic stabilization. We now turn to the computation of an optimal rule. The foundations of an optimal rule (as exemplified by [Woodford \(2003\)](#) for instance for monetary policy) can be linked to the idea that policymakers adopt policy measures aimed at maximising social welfare in the economy. The literature offers alternative ways of defining the social welfare (based either on the maximisation of the intertemporal utility of representative consumers or, as a by product, on the minimisation of a loss function). Following [Woodford \(2003\)](#), the approximation up to second order to the welfare reads as (see appendix for the maths):

$$\mathcal{W}_t \simeq \frac{1}{1-\beta} \left(\bar{U} - \frac{1}{2} [\kappa \text{var}(\pi_t) + (\sigma_C + \chi \sigma_H) \text{var}(Y_t)] \right). \quad (24)$$

Abstracting from the steady state, the welfare function may be rewritten as a loss function:

$$\mathcal{L}_t \simeq -\mathcal{W}_t \quad (25)$$

$$= \lambda^\pi \text{var}(\hat{\pi}_t) + \lambda^y \text{var}(\hat{y}_t) + \lambda^r \text{var}(\hat{r}_t). \quad (26)$$

In this expression, each parameter quantifies welfare losses experienced by households (or policymakers) when inflation, output gap, interest rate volatilities increase.

Concerning monetary authorities, they choose the value of the interest rate rule (i.e., parameters ϕ_π and ρ) on the minimisation of this loss function, subject to the structure of the economy. More technically the last weight of the loss function, λ^r , is a parameter that accounts for the zero lower bound probability. Indeed, paying negative interest rate on the monetary base is technically infeasible and represents a constraint for the implementation of an optimal policy. In normal times with low shocks, this constraint is not binding (i.e., as $\hat{r}_t > 0$, $\lambda^r = 0$). However, following the recent financial turmoil, characterised by large shocks, refinancing interest rates hit the zero lower bound in some countries thus forcing central banks to launch unconventional measures. Moreover, this parameter also accounts for the loss experienced by the economy when the central bank adjust too much its interest rate.

Regarding weights on the variance of output λ^y and inflation λ^π , we can deduct them from the welfare function:

$$\lambda^\pi = \frac{0.5\kappa}{1-\beta}, \quad \lambda^y = \frac{0.5(\sigma_C + \chi\sigma_H)}{1-\beta}. \quad (27)$$

According to our loss function, a first best situation is characterized by a stable inflation rate, an output gap under control and stable monetary policy. If we minimise the loss function under the economy constraint, our economy is expected to approach this first best situation. The problem faced by policymakers when implementing an optimal fiscal and monetary policy reads as follows:

$$(\rho, \phi_\pi, \phi_Y) = \arg \min_{\rho, \phi_\pi, \phi_Y} \mathcal{L}$$

The computation of the optimized parameters is straightforward as the objective function (*i.e.* the loss function) is quadratic under a linear constraint (*i.e.* our economy). Dynare is able to solve this type of optimisation problem. All we need to do is to specify to dynare the shape of the loss function and the coefficients to use in the optimisation exercise.

Let us first declare the loss function using the `optim_weights` blocks. Each line, we have to provide the name of the endogenous variable (whose variance will be minimised) and its associated weight. The conversion of the loss function 25 in Dynare's syntax delivers the following code:

```
gamm_pi    = .5/(1-beta)*kappa;
gamm_y     = .5/(1-beta)*(sigmaC+(epsilon-1)/epsilon*sigmaH);
optim_weights;
pi gamm_pi;
y gamm_y;
end;
```

We also use to specify which parameters are going to be optimised. Focusing on monetary policy, we optimise the two coefficients in the taylor rule: ρ, ϕ_π . The relevant Dynare syntax is:

```
osr_params rho phi_pi phi_y;
```

Finally, to run the optimisation and the draw of IRFs under the optimal policy, the `osr` command is required. This function works exactly as `stoch_simul`, the only difference is that `osr` compute the optimal policy before performing simulations.

```
osr ly lp lr;
```

Running our code, we obtain the following response from the model:

console response

```
OPTIMAL SIMPLE RULE
OSR: Initial value of the objective function: 1.21389
-----
f at the beginning of new iteration, 1.2138857611
Predicted improvement: 0.000414694
lambda = 1; f = 1.2130568
lambda = 1.9332; f = 1.2122839
lambda = 3.7372; f = 1.2107918
lambda = 7.2247; f = 1.2079146
lambda = 13.967; f = 1.2023799
lambda = 27; f = 1.1917820
lambda = 52.196; f = 1.1716675
lambda = 100.9; f = 1.1341306
lambda = 195.07; f = 1.0662892
lambda = 377.1; f = 0.9507665
lambda = 729; f = 0.7739818
lambda = 1409.3; f = 0.5478851
Norm of dx 0.00028799
```

....

The social planner is now looking for the best parameters, *i.e.* parameters that minimise the loss of the economy (or the weighted volatility). After 22 iterations, the algorithm stops as the last iteration efficiency was too small. The optimal loss is now 3.9 instead of 5.1.

console response

```
Improvement on iteration 69 = 0.000000052
improvement < crit termination
OPTIMAL VALUE OF THE PARAMETERS:
rho 0.991898
phi_r 163.631
Objective function : 3.91181
```

After, Dynare also reveals the value of the parameters which minimises the social loss of the economy:

console response

OPTIMAL VALUE OF THE PARAMETERS:

rho 0.991898

phi_r 163.631

Objective function : 3.91181

The model suggests that monetary policy should be highly reactive to inflation $\phi_\pi = 114$ with a lot of inertia $\rho = 0.97$. This result is standard with the use of loss functions, see for instance [Darracq-Pariès et al. \(2011\)](#). Using this coefficient, we are able to see how the economy would have performed under an optimal rule, how an optimal policy improves the volatility of an aggregate at the expense of another. We are also able to do a ranking of different policies, and explain why one policy is more suitable than another. Analysing optimal policy under different types of shocks, so that policymakers can adjust the policy according to the type of shock hitting the economy (for instance, monetary should concentrate on markup and demand shocks while it should not target productivity shocks). Our set-up is also amenable for analysing the interaction between monetary and fiscal policy.

Exercise 1: The determinants of monetary policy design

1. Determine which shocks play a role on the welfare function, and thus on the reaction from monetary policy. How are the coefficients affected when you disabled some of the shocks? Illustrate the “divine coincidence” of [Blanchard and Galí \(2007\)](#).

	All shocks	Productivity shocks only	Preference shocks only	Supply shocks only
ρ	...	...	...	...
ϕ_π	...	...	...	...
ϕ_y	...	...	...	...

2. Set a value of the adjustment cost on prices close to zero. What are the implications for optimal monetary policy?

	Benchmark $\kappa = 150$	Small price costs $\kappa = \dots$
ρ	...	...
ϕ_π	...	...
ϕ_y	...	...

3. As in [Lucas Jr \(2003\)](#), we are going to compute the welfare cost of business cycles. To evaluate the cost of business cycles, we are going to compute how much households would be willing to pay to live in an economy with no fluctuations. The no-arbitrage condition is given by:

$$\bar{W} = \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ \frac{1}{1 - \sigma_C} [(1 + \lambda) C_{t+\tau}]^{1 - \sigma_C} - \frac{\chi}{1 + \sigma_H} H_{t+\tau}^{1 + \sigma_H} \right\},$$

where $\bar{W}$ is the steady state level of welfare and λ is the welfare cost. The left hand side is the welfare in absence of fluctuations (at steady state) and the right hand side is the regime with fluctuations. We thus determine λ , the fraction of consumption that the households would need to be indifferent between these two policy regimes. Applying the second order

approximation:

$$\bar{\mathcal{W}} = \frac{1}{1-\beta} \left((1+\lambda)^{1-\sigma_C} \left[\bar{f} + \frac{\bar{f}_{yy}''}{2} \text{var}(Y_t) + \frac{\bar{f}_{\pi\pi}''}{2} \text{var}(\pi_t) \right] - \left(\bar{g} + \frac{\bar{g}_{yy}''}{2} \text{var}(Y_t) \right) \right)$$

Isolating the welfare cost reads as follows:

$$\lambda = \left(\frac{\left[(1-\sigma_C)^{-1} + \frac{1}{2} \frac{(\epsilon-1)}{\epsilon} \sigma_H \text{var}(Y_t) \right]}{\left[(1-\sigma_C)^{-1} - \frac{\sigma_C}{2} \text{var}(Y_t) - \frac{\kappa}{2} \text{var}(\pi_t) \right]} \right)^{1/(1-\sigma_C)} - 1$$

- From `osr()` output, get the variance of output and inflation from matrix `oo_.var`.
- Compute λ at the end of the Dynare code.
- Compare the cost of fluctuations under an optimal vs the calibrated rule. How has increased the welfare in terms of consumption between these two policy regimes?

Exercise 2: The determinants of monetary policy design

Optimal monetary policy faces of trade-off between stabilization of output and inflation. To illustrate the trade-off.

1. Normalize to one the weight on inflation $\lambda^\pi = 1$.
2. Run a loop that iterate on different values for the weight on output $\lambda^\pi = 0 : 0.01 : 1$. For each value, compute the optimal policy, given this change in the weight on output.
3. From `osr()` function output, store the both standard deviations of inflation and output for each iteration of the loop.
4. Plot the these variances into a plan $(\text{var}(\pi), \text{var}(y))$. Do you see any trade-off?

Exercise 3: Estimating the model

Estimate the model using the data from the previous handout.

References

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Appendices

4. Micro-founding the social welfare

The use of a loss function has many advantages for policy makers as it allows a perfect control over the objectives achieved by macroeconomic policies. However, the objectives as defined in the weights of the loss function don't rely on the utility function of the representative households, such that an optimal policy as defined by an ad hoc loss function may be harmful in terms of social welfare. [Woodford \(2003\)](#) provides a path-breaking contribution regarding the microfoundation of optimal policies by derivating a loss function obtained making an approximation of the welfare criterion of households. In this appendices, I provide a similar exercise as in [Woodford \(2003\)](#) in a more simpler framework to illustrate how a loss function can be obtained from a second order approximation of the utility function.

4.1 A second order approximation

Here, I am making a second order approximation of the utility function. I am not going to approximation up to second order all the equations, I will only focus on equations which really matters regarding the optimal conduct of optimal monetary policy.²

4.1.1 Stochastic calculus up to second order

Let us first provide a second order taylor expansion to any function, with x_t a iid gaussian random variable, the second order approximation of $f(x_t)$ reads as:

$$f(x_t) \simeq f(\bar{x}) + f'_x(\bar{x})(x_t - \bar{x}) + \frac{1}{2}f''_{xx}(\bar{x})(x_t - \bar{x})^2 + ||O_3||$$

²The derivation by hand of a second order welfare criterion is most of the time arbitrary and may lead to fallacious results. [Schmitt-Grohé and Uribe \(2004\)](#) develops an accurate alternative by developping a second order approximation of all equilibrium conditions of the model (instead of "most important" conditions).

where $||O_3||$ is the higher order term (that we neglect). The unconditional mean of this function is given by:

$$\begin{aligned} E[f(x_t)] &\simeq f(\bar{x}) + f'_x(\bar{x}) E[(x_t - \bar{x})] + \frac{1}{2} f''_{xx}(\bar{x}) E[(x_t - \bar{x})^2] + ||O_3|| \\ &\simeq f(\bar{x}) + \frac{1}{2} f''_{xx}(\bar{x}) \text{var}(x_t) \end{aligned}$$

Here, because of the gaussian assumption for the distribution of x_t , on average the linear deviations of a gaussian variable from its mean are asymptotically 0: $E[(x_t - \bar{x})] \simeq 0$. Put differently, since $x_t \sim N(\bar{x}, \text{var}(x_t))$, then $E[x_t] = \bar{x}$.

4.1.2 An application to the welfare criterion

The welfare function reads as:

$$\mathcal{W}_t = \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ \frac{1}{1 - \sigma_C} C_{t+\tau}^{1-\sigma_C} - \frac{\chi}{1 + \sigma_H} H_{t+\tau}^{1+\sigma_H} \right\},$$

where the left term of the utility function denotes the utility from consumption goods, while the right term is the disutility cost of producing goods. An optimal policy aims at increasing the amount of goods consumed and diminishing the utility cost of producing them.

The goal at this point is to express our welfare into output and inflation terms. Substituting the consumption by the resources constraint: $C_t = Y_t (1 - 0.5\kappa(\pi_t - 1)^2)$, and hours worked by the production function, $H_t = Y_t / \varepsilon_t^A$, then the welfare reads as:

$$\begin{aligned} \mathcal{W}_t &= \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ \frac{1}{1 - \sigma_C} \left(Y_{t+\tau} \left(1 - \frac{\kappa}{2} (\pi_{t+\tau} - 1)^2 \right) \right)^{1-\sigma_C} - \frac{\chi}{1 + \sigma_H} \left(\frac{Y_{t+\tau}}{\varepsilon_{t+\tau}^A} \right)^{1+\sigma_H} \right\} \\ &= \mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \left\{ f(Y_{t+\tau}, \pi_{t+\tau}) - g(Y_{t+\tau}, \varepsilon_{t+\tau}^A) \right\} \end{aligned}$$

First, regarding the steady state, recall that:

$$\begin{aligned} \bar{Y} &= \bar{C} = \bar{H} = \bar{\pi} = 1 \\ \chi &= \frac{(\epsilon - 1)}{\bar{H}^{1+\varphi_\epsilon}} = \frac{(\epsilon - 1)}{\epsilon} \end{aligned}$$

Then, using a second order expansion:

$$f(Y_t, \pi_t) \simeq \bar{f} + \bar{f}'_y (Y_t - \bar{Y}) + \frac{\bar{f}''_{yy}}{2} (Y_t - \bar{Y})^2 + \bar{f}'_\pi (\pi_t - \bar{\pi}) + \frac{\bar{f}''_{\pi\pi}}{2} (\pi_t - \bar{\pi})^2 + ||O_3||$$

Then, the left term of the welfare criterion can be approximated by:

$$\mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \{f(Y_{t+\tau}, \pi_{t+\tau})\} \simeq \frac{1}{1 - \beta} \left(\bar{f} + \frac{\bar{f}''_{yy}}{2} \text{var}(Y_t) + \frac{\bar{f}''_{\pi\pi}}{2} \text{var}(\pi_t) \right) + ||O_3|| + t.i.p.$$

where here, *t.i.p.* are terms which are independent of the policy (here ε_t^C is an exogenous shock that cannot be affected by monetary policy). These are affecting the welfare criterion, but cannot be affected by the policy. Then for clarity purposes, there is no need to write them down in the welfare criterion.

In the same vein, the second term reads as:

$$\mathbb{E}_t \sum_{\tau=0}^{\infty} \varepsilon_{t+\tau}^C \beta^\tau \{g(Y_{t+\tau}, \varepsilon_{t+\tau}^A)\} \simeq \frac{1}{1-\beta} \left(\bar{g} + \frac{\bar{g}_{yy}''}{2} \text{var}(Y_t) \right) + ||O_3|| + t.i.p.$$

In the latter expression, both shocks are included in the t.i.p. term.

To summarize, the welfare criterion now reads as:

$$\begin{aligned} \mathcal{W}_t &\simeq \frac{1}{1-\beta} \left(\bar{f} + \frac{\bar{f}_{yy}''}{2} \text{var}(Y_t) + \frac{\bar{f}_{\pi\pi}''}{2} \text{var}(\pi_t) - \left(\bar{g} + \frac{\bar{g}_{yy}''}{2} \text{var}(Y_t) \right) \right) + ||O_3|| + t.i.p. \\ &\simeq \frac{1}{1-\beta} \left(\bar{U} + \frac{1}{2} [(\bar{f}_{yy}'' - \bar{g}_{yy}'') \text{var}(Y_t) + \bar{f}_{\pi\pi}'' \text{var}(\pi_t)] \right) + ||O_3|| + t.i.p. \end{aligned}$$

We now need to find $\bar{f}_{yy}''$, $\bar{g}_{yy}''$ and $\bar{f}_{\pi\pi}''$. Given the very simple steady state of the model, these derivatives boils down to this:

$$\begin{aligned} \bar{f}_{yy}'' &= -\sigma_C \\ \bar{f}_{\pi\pi}'' &= -\kappa \\ \bar{g}_{yy}'' &= \chi\sigma_H = \frac{(\epsilon-1)}{\epsilon}\sigma_H \end{aligned}$$

Finally, the welfare function reads as:

$$\mathcal{W}_t \simeq \frac{1}{1-\beta} \left(\bar{U} - \frac{1}{2} [\kappa \text{var}(\pi_t) + (\sigma_C + \chi\sigma_H) \text{var}(Y_t)] \right).$$

This equation is key here to determine the social cost of inflation. The level of goods in the economy consumed by the household decrease in the stickiness of prices $\kappa \text{var}(\pi_t)$. A sticky economy $\kappa > 0$ experiences a cost of inflation that a monetary policy aims at stabilizing through the nominal interest, while in a flexible price economy, $\kappa = 0$, the allocation of consumption is optimal as there is no cost of nominal fluctuations. In addition, the standard Lucas cost of business cycles is proportional to the risk aversion coefficient σ_C , while the cost of supplying labour is proportional to $\chi\sigma_H$.